PHD PROGRESS REVIEW · OCTOBER 2026

early-psycho

Whole-brain models of early psychosis: what the first iteration showed, what we fixed, and what we are testing now

HCP for Early Psychosis · 48 HC · 25 AP · 76 NAP

Hopf / GEC · in-silico perturbation · O-information

github.com/fedemoss/early-psycho

ROADMAP

Four parts

01 · v1

First iteration

GEC per subject, perturbability, P2H / P2O and a family of PANSS models. Headline result: $R^2 = 0.31$.

02 · audit

What we fixed

Feature-selection leakage, model and analysis bugs, an inverted P2H sign, a full refit.

03 · v2

New hypotheses

Hierarchy and irreversibility (Acero-Pousa), Mana et al. priors, higher-order interactions, leak-free prediction.

04 · next

Next steps

Robustness of the irreversibility effect, richer clinical data, HOI extensions, papers to read.

DATA

HCP for Early Psychosis, resting state

Group	n	Clinical scores
Healthy controls (HC)	48	none
Affective psychosis (AP)	25	PANSS total, YMRS
Non-affective psychosis (NAP)	76	PANSS total, YMRS

PANSS available for 92 patients, YMRS for 98. MADRS is empty in our files, and the timeseries arrays carry no subject IDs (the raw Schaefer-1000 runs do).

Acquisition

- Schaefer-100, 7 Yeo networks (cortex only)
- TR = 0.8 s, 1581 frames (4 runs)
- Band-pass 0.008–0.08 Hz
- Group-average structural connectome

Jacobs et al. 2024, Schizophr Bull

01

First iteration (v1)

The legacy pipeline (May–Sep 2026): what we computed and what it seemed to show.

V1 · PIPELINE

From BOLD to effective connectivity to symptoms

rs-fMRI
149 subjects, Schaefer-100

→

FC & lag-1 FS
zero-lag and time-shifted correlations

→

Hopf model
GEC fitted per subject from the SC

→

Perturbation
FDT perturbability, P2H, P2O

→

Symptoms
PANSS models

GEC

Linear Hopf ($a = -0.02$), pseudo-gradient fit to FC + FS($\tau = 1$ TR). Fit $r(FC) \approx 0.61$.

P2H / P2O

Inject extra noise at node j; measure how the simulated FC moves toward the healthy mean (P2H) or away from baseline (P2O).

Perturbability

FDT-based response map per node (Deco / Kringelbach). Also GEC asymmetry $\|C - C^T\|$.

legacy/gec_and_pert_analysis.py · legacy/notebooks/gec_matrices.ipynb

V1 · PANSS MODELS

Many models, one apparent winner

Model (all LOOCV)	R^2	r	Note
XGBoost on row/column summaries of GEC, FC, FS	−0.27	–	3-class version: 52% acc. (majority 59%)
XGBoost “big” models (FC+FS / GEC / all)	−0.24 … −0.01	–	nothing learned
Raw Schaefer-1000 model	0.07	–	subjects duplicated (AP/PA phase files)
SC-selected edges + ridge	−0.02	–	Jaccard with corr-selected set: 0.006
lag-1 FC edges, |corr(edge, PANSS)| > 0.2 + ridge	**0.31**	**0.56**	233 / 9900 edges
Same with GEC edges	0.22	0.47	133 edges
+ P2H / P2O as candidate features	0.31	0.56	none passed selection

legacy/notebooks/clinical_scores_*.ipynb · panss_model_region_overfit.ipynb · panss_model_sc_regions.ipynb

Perturbation measures did not track PANSS

Direct correlation

max(P2O) vs PANSS: r = 0.09 (p = 0.38)
min(P2H) vs PANSS: r = −0.13 (p = 0.22)

Same null for P2H/P2O vs the model's per-subject error.

Perturb → predicted PANSS

Node minimising predicted PANSS = node minimising P2H in **0 / 92** subjects. Weak rank effect (44.7 vs 50.5, p = 0.004).

Used the leaky model, and TR = 2.0 instead of 0.8.

One region everywhere

`argmin(P2H)` = region 26 (LH salience lateral PFC) in 78 / 92 patients **and** in 40 / 48 controls.

Group effects unstable

GEC asymmetry: HC < AP in the 10-subject pilot (0.052 vs 0.061), HC > AP in the full run (0.064 vs 0.049).

V1 · TAKE-AWAY AT THE TIME

$R^2 = 0.31$

A ridge model on 233 lag-1 FC edges seemed to predict PANSS (r = 0.56).
Perturbation measures added nothing. The next question was: is that signal real?

02

Audit and fixes

Leakage in the headline model, bugs in the model port, an inverted sign, and a full refit.

AUDIT · THE $R^2 = 0.31$ MODEL

The headline result was feature-selection leakage

Edges selected on all 92 subjects (v1)

0.31

LOOCV R^2, the held-out subject helped pick the features

Selection redone inside each fold

−0.21

−0.21 to −0.33 across thresholds 0.15–0.25

Leaky pipeline on shuffled PANSS

0.15

mean of 200 shuffles; **11%** reach ≥ 0.31 ($p \approx 0.11$)

With 9,900 candidate edges and 92 subjects, full-sample screening produces positive CV performance on pure noise. From now on: selection and tuning inside folds, plus a permutation null of the whole pipeline.

src/symptom_prediction/leakage_audit.py · Varoquaux 2018

AUDIT · THE MODEL PORT

Six issues in the v1 Python pipeline

Issue	v1	v2 fix
Oscillation frequency	Jacobian used f (Hz)	$\omega = 2\pi \cdot f$, as in the papers
Coupling scale	GEC rescaled to max 1	max 0.1, as in `Ceff_EP_HCP.m`
Lagged target	empirical FS transposed vs model	both $\mathrm{cov}(x_i(t), x_j(t+\tau))$
Homotopic links	region i ↔ i+50 (not homologues in Schaefer)	mirrored centroids, same network
TR	2.0 s in the perturbation→PANSS notebook	0.8 s everywhere
Clinical model evaluation	full-sample selection	nested CV, within-group permutation null

src/core/hopf.py · src/core/evaluation.py

AUDIT · P2H

The “best” perturbation was the most damaging one

$$P2H_j = \mathrm{sim}(FC_{\mathrm{pert}\ j}, FC_{HC}) - \mathrm{sim}(FC_{base}, FC_{HC})$$

- Higher P2H = the perturbation moves the subject **toward** healthy
- v1 notebooks used `argmin(P2H)` as “closest to healthy”: that is the node whose perturbation **hurts** similarity most
- P2H < 0 for essentially every node and subject: no single-node noise injection improves anyone
- Region 26 won in controls too, so it reflects the model and atlas, not the patients

v2 conventions

best node = argmax

Healthy reference computed leave-one-out for controls. Added Mana-style perturbations of the bifurcation parameter a_j.

REFIT

The fixed model fits much better

v1 fit, r(FC)

0.61

r(FS) 0.59

→

v2 fit, r(FC)

0.77

r(FS) 0.75

Group	r(FC) v1	r(FC) v2
HC	0.62	0.77
AP	0.62	0.78
NAP	0.60	0.77

All 149 subjects refitted (≈ 20 min on 8 cores, locally). Fit quality is similar across groups, so group differences are not a fit artefact.

03

New hypotheses (v2)

Literature-driven questions, all tested with the fixed model and leak-free statistics.

V2 · HYPOTHESES

Four questions from the literature

H1 · hierarchy

More non-equilibrium in patients

Acero-Pousa 2025: higher FDT deviation (“hierarchy”) in chronic schizophrenia. Same in early psychosis? Plus model-free irreversibility.

Not supported: no FDT difference; irreversibility goes the opposite way (lower).

H2 · Mana priors

Restorative regions rank high

Mana 2023: perturbing OFC, posterior cingulate, parahippocampal cortex restores healthy dynamics. Top of our P2H maps?

Not supported: Mana sets sit at chance in every map.

H3 · higher-order

Less synergy, more redundancy

Tarchi 2026: schizophrenia loses synergy. Same here, and can the Hopf model reproduce it?

Not supported: non-significant trend the other way; the model makes no synergy.

H4 · symptoms

Features predict symptoms

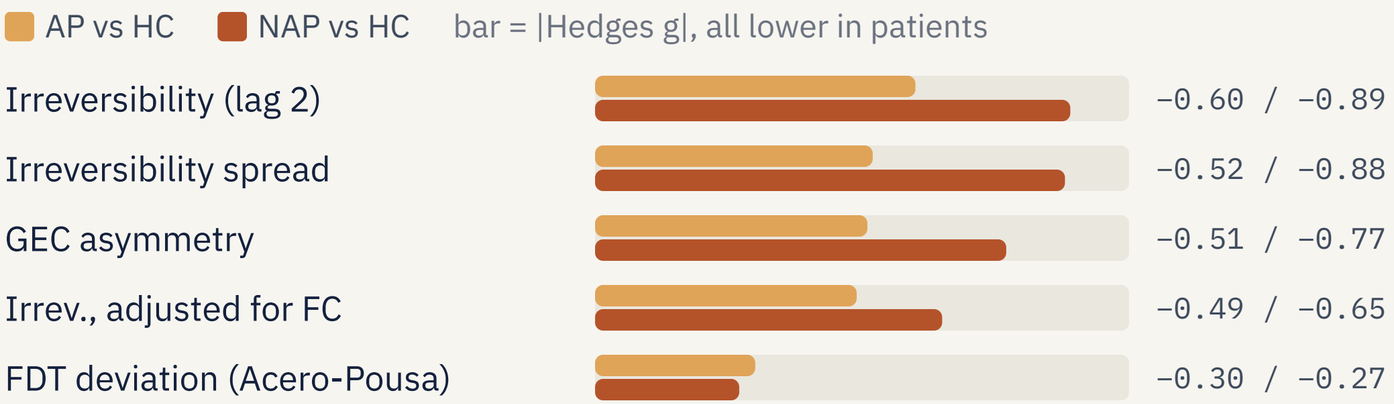
Do any features, or literature priors (Keane 2025), predict PANSS or YMRS beyond the AP/NAP difference?

Not supported: all 13 feature sets at permutation-null level.

references on the last slide

H1 · RESULT

Patients are less time-irreversible than controls



VERDICT · H1 NOT SUPPORTED (AS STATED)

The Acero-Pousa measure shows no group difference, and the model-free effect runs the **opposite** way: patients' dynamics are closer to equilibrium, not further. The reversed effect itself is robust and is now our main finding.

Irreversibility: FDR q < 0.001, at lags 1–3 and in **all 7 networks** (NAP g −0.79 to −0.90).

Model asymmetry tracks empirical irreversibility (ρ = 0.91). FDT deviation: no group difference (p = 0.27).

Opposite direction to Acero-Pousa (chronic SZ). No symptom correlation survives FDR.

src/hierarchy/run.py · notebooks/02_hierarchy.ipynb · Acero-Pousa 2025

H2 · RESULT

Mana et al. regions are not enriched in our P2H maps

Test (mean within-subject percentile)	Observed	p vs random sets
Restorative set, P2H noise, patients	0.52	0.34
Restorative set, P2H $a_j \downarrow$, patients	0.52	0.26
Pathogenic set, P2P $a_j \uparrow$, controls	0.51	0.40

Mana’s Desikan-Killiany regions mapped to Schaefer via Harvard-Oxford overlap: 12 restorative nodes (OFC / vPFC, posterior cingulate, parahippocampal) and 18 pathogenic nodes (postcentral, insula, inferior parietal, fusiform, MTG).

VERDICT · H2 NOT SUPPORTED

Neither set ranks above random nodes. Caveat: a loose replication (different atlas, noise-injection readout instead of Mana’s metastable-state KL divergence).

Most frequent best P2H node (patients)

LH precuneus / PCC

48 / 92

Adjacent to Mana’s posterior cingulate but just under the overlap threshold. Post hoc, suggestive only.

src/perturbation/run.py · notebooks/03_perturbation.ipynb · Mana et al. 2023

H3 · RESULT

Higher-order: a trend in the data, and a model that cannot make synergy

Empirical O-information (161,700 triplets)

Measure	AP vs HC	NAP vs HC	p
Mean Ω (redundancy)	−0.44	−0.36	0.07
Fraction synergistic	+0.41	+0.35	0.10

Patients trend less redundant, more synergistic; partly overlaps their lower mean FC. No PANSS / YMRS correlation.

GEC model vs data

empirical (about) **25%**

GEC model **0%**

of triplets are synergy-dominated. Within-subject $r(\Omega_{emp}, \Omega_{model}) \approx 0.66$. Non-negative pairwise couplings cannot produce synergy (Caprioglio et al. 2026).

VERDICT · H3 NOT SUPPORTED

No significant group difference, and the trend points to **more** synergy in patients, the opposite of Tarchi 2026. The Hopf model cannot test it: with non-negative couplings it produces no synergy.

src/higher_order/run.py · notebooks/04_higher_order.ipynb · Rosas 2019

H4 · RESULT

No feature set predicts PANSS or YMRS once evaluation is leak-free

Feature set (6 of 13 shown)	PANSS p	YMRS p
FC edges, in-fold top-200	0.52	0.16
GEC edges (ridge, all)	0.70	0.56
O-information network compositions	0.79	0.75
FDT P_j + nodal irreversibility	0.68	0.34
Mana restorative P2H	0.38	0.77
Keane 2025 SomMot–Vis edges	0.44	0.53

VERDICT · H4 NOT SUPPORTED

Every feature set is at permutation-null level (all p ≥ 0.16). PANSS total and YMRS may be the wrong targets.

Protocol

Repeated 10-fold CV; selection, scaling and ridge α inside folds; AP/NAP means removed in-fold; 200-permutation null of the full pipeline, shuffling within group.

R^2 never exceeds the group-only model. Consistent with Foster 2025: PANSS **total** is not predictable in early psychosis, but its factors were.

src/symptom_prediction/run.py · notebooks/05_symptom_prediction.ipynb · Foster et al. 2025

04

Where we stand, and next steps

One robust group effect, honest nulls for symptoms, and a clear list of things to do.

SUMMARY

What we know now

robust

Lower irreversibility in psychosis

g ≈ −0.9 (NAP), −0.6 (AP); every network and lag; survives FC adjustment; mirrored by GEC asymmetry.

improved

A correct, better-fitting model

Six fixes; fit 0.61 → 0.77; code reorganised in `src/` with notebooks per study.

trend

Less redundancy, more synergy

p ≈ 0.07–0.10; the linear model produces no synergy at all.

null

No symptom prediction, no Mana enrichment

The v1 $R^2 = 0.31$ was leakage; PANSS total and YMRS not predictable; Mana sets at chance.

v2 pipeline rerun from src/ on 22 Oct; all results reproduced

NEXT STEPS

To do, in priority order

1 · done (22 Oct)

Rerun v2 and push figures

All results reproduced from `src/`; notebooks 01–05 executed with figures.

2 · irreversibility

Stress-test the main effect

Motion (FD), CPZ dose, global signal, per-run vs concatenated, τ; FDT at τ = 2 TR.

3 · clinical data

NDA #2914 tables

Item-level PANSS (Wallwork factors), CAINS / BNSS, MADRS, age, sex; ID-aligned rebuild.

4 · parcellation

Add subcortex and cerebellum

Thalamo-cortical (Keane) and cerebellar–PFC (Brady) circuits need these parcels.

5 · higher-order

Dynamics and a signed model

Lagged O-information; GEC with negative couplings; perturbation-to-healthy in O-info space.

6 · Mana

Closer replication

Their readout (KL over metastable substates), DK-matched regions, symptom factors.

REFERENCES

Papers behind each part

Data, models, evaluation

- Jacobs et al. 2024 · HCP for Early Psychosis · Schizophr Bull
- Deco et al. 2023 · FDT violation and perturbability · Phys Rev E
- Acero-Pousa et al. 2025 · hierarchy in schizophrenia · Transl Psychiatry
- Mana et al. 2023 · in-silico perturbation in SZ · Cereb Cortex
- Varoquaux 2018 · small-sample CV error bars · NeuroImage
- Foster et al. 2025 · CPM of psychosis symptoms · Neuropsychopharmacology
- Keane et al. 2025 · somato-visual dysconnectivity · Mol Psychiatry
- Wallwork et al. 2012 · PANSS five-factor model · Schizophr Res

Higher-order interactions

- Rosas et al. 2019 · O-information · Phys Rev E
- Luppi et al. 2022 · synergistic core · Nat Neurosci
- Tarchi et al. 2026 · ΦID in schizophrenia · Brain Behav
- Caprioglio et al. 2026 · synergistic motifs in Gaussian systems · PRL
- Gatica et al. 2022 · HOI in a whole-brain model · PLoS Comput Biol
- Herzog et al. 2024 · ketamine HOI · Transl Psychiatry
- Stramaglia et al. 2021 · dynamical O-information · Front Physiol
- Santoro et al. 2024 · higher-order connectomics · Nat Commun

code: github.com/fedemoss/early-psycho